Day 20 . 9/11/26

21 - Merge Two Sorted Lists

Approach

list1 = 1 → 3 → 5

list2 = 2 → 4 → 6

Time Complexity
O(n + m)
Space Complexity
O(1)

We maintain a current pointer.

1. Compare list1 and list2.
2. Take the smaller node.
3. Attach it to the merged list.
4. Move the list's pointer forward.
5. Continue until one list becomes null.
6. Attach the remaining part of the other list.

Code :-

```
class Solution {
    public ListNode mergeTwoLists(ListNode list1, ListNode list2){
        ListNode dummy = new ListNode(0);
        ListNode current = dummy;
        while (list1 != null && list2 != null) {
            if(list1.val <= list2.val){
                current.next = list1;
                list1 = list1.next;
            }else{
                current.next = list2;
                list2 = list2.next; }
            current = current.next; }
        if (list1 != null) current.next = list1;
        else current.next = list2;
        return dummy.next;
    }
}
```

Description | Accepted | Editorial | Solutions | Submr

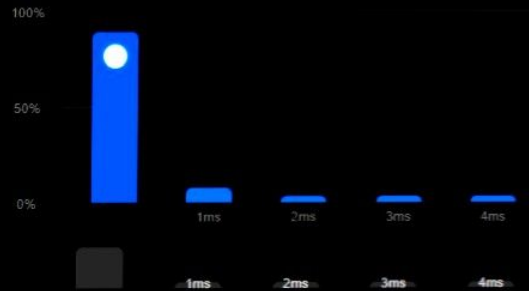
All Submissions

Runtime

0 ms | Beats **100.00%**

Memory

43.88 MB | Beats **76.93%**



Code Java

Code

Java | Auto

```java
public ListNode mergeTwoLists(ListNode list1, ListNode list2) {

    ListNode dumy = new ListNode(0);
    ListNode curr = dumy;
    while(list1 != null && list2 != null){
        if(list1.val  <= list2.val){
            curr.next = list1;
            list1 = list1.next;
        }else{
            curr.next=list2;
            list2 = list2.next;
        }
        curr = curr.next;
    }
    if(list1 != null){
        curr.next = list1;
    }else{
        curr.next = list2;
    }
    return dumy.next;
```

Saved

Ln 15, Col 30

Testcase | Test Result

Accepted Runtime: 0 ms